

V. CONCLUSION

In this paper, we have shown that there are advantages in formulating observers as DAE's which can be implemented using existing DAE solvers. A design method based on the extended linearization approach is presented.

There are many possible extensions to the work presented here. An immediate one is to allow the measurement equation to depend on the input u , i.e., have $y = h(x, u)$ instead of (2) (this situation seldom comes up in real applications). All the results presented here can be easily extended to this case. A more interesting extension is to consider the observer problem with unknown inputs. Unknown inputs in a sense are relaxation variables, and the problem of estimation with unknown inputs falls nicely into the DAE framework. The difficulty in formulating the DAE observer in that the case lies in proper combination of the unknown inputs and the (reduced) set of relaxation variables (λ 's) so that the resulting DAE has the required properties. The observer with unknown inputs has applications in residual generation for failure detection. In fact, the relaxation variable λ can be directly interpreted as a residual so that in this case the canonical DAE observer can be considered as a *residual generator*.

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Sliding Mode Control Design Based on Ackermann's Formula

Juergen Ackermann and Vadim Utkin

Abstract—The sliding mode control methods are developed to design systems which have the desired dynamic behavior and are robust with respect to perturbations. It is shown that the discontinuity plane for sliding mode control may be found in an explicit form using Ackermann's formula. Two design procedures are derived. First, static controllers are designed to enforce sliding modes with the desired dynamic properties after a finite-time interval. Then, dynamic controllers are designed that exhibit the desired dynamic properties during the entire control process.

Index Terms—Ackermann's formula, linear system, sliding mode.

I. INTRODUCTION

Ackermann's formula enables one to obtain a linear state-feedback control law in an explicit form resulting in a feedback system with desired eigenvalues [1]. A similar task arises when designing sliding mode control in linear systems with a linear discontinuity surface since the corresponding sliding mode equation is linear and depends on the coefficients of the surface equation [2].

For an arbitrary linear control system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{D}\mathbf{f}(t)$$

with state, control, and disturbance vectors $\mathbf{x}(t) \in \mathbb{R}^n$, $\mathbf{u}(t) \in \mathbb{R}^m$, and $\mathbf{f}(t)$, and $\text{rank}(\mathbf{B}) = m$, sliding mode control is designed as a state function undergoing discontinuities in manifold $\mathbf{s} = \mathbf{C}\mathbf{x} = 0$ ($\mathbf{C}$ is constant, $\mathbf{s} \in \mathbb{R}^m$) such that starting from some finite-time instant the state trajectories are confined to this manifold. Then, the sliding motion is governed by the reduced-order system and it does not depend on the disturbance if the matching condition $\mathbf{D} \in \text{span}(\mathbf{B})$ is satisfied [3]. To derive the reduced-order motion equation, the m state variable may be excluded using the algebraic system $\mathbf{s} = 0$, and control $\mathbf{u}$ should be substituted by the so-called "equivalent control" that is the solution to $\dot{\mathbf{s}} = 0$ with respect to the control [2].

The traditional approach implies transforming the linear system into the phase canonic form which is a particular case of the so-called regular form [4]–[6] consisting of two blocks

$$\dot{\mathbf{x}}_1 = \mathbf{A}_{11}\mathbf{x}_1 + \mathbf{A}_{12}\mathbf{x}_2$$

$$\dot{\mathbf{x}}_2 = \mathbf{A}_{21}\mathbf{x}_1 + \mathbf{A}_{22}\mathbf{x}_2 + \mathbf{B}_2\mathbf{u}$$

where $\mathbf{x}_1 \in \mathbb{R}^{n-m}$, $\mathbf{x}_2 \in \mathbb{R}^m$, and $\det(\mathbf{B}_2) \neq 0$. The first block does not depend on the control, while the dimension of the second block is equal to that of the control.

Then $\mathbf{x}_2$ is handled as a control in the first block and designed as a linear function of $\mathbf{x}_1$

$$\mathbf{x}_2 = -\mathbf{C}_1\mathbf{x}_1$$

($\mathbf{C}_1$ is constant) in accordance with some performance criterion. Control is selected as a discontinuous function of the state

$$\mathbf{u} = \begin{cases} \mathbf{u}^+(\mathbf{x}), & \text{if } \mathbf{s} > 0 \\ \mathbf{u}^-(\mathbf{x}), & \text{if } \mathbf{s} < 0 \end{cases} \quad \text{component-wise} \quad (1)$$

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J. Ackermann is with DLR, German Aerospace Research Establishment, Institute for Robotics and System Dynamics, 82230 Wessling, Germany.

V. Utkin is with the Department of Electrical Engineering, The Ohio State University, Columbus, OH 43210 USA (e-mail: utkin@ee.eng.ohio-state.edu). Publisher Item Identifier S 0018-9286(98)00927-1.

(where $s = \mathbf{C}_1 \mathbf{x}_1 + \mathbf{x}_2$ and $\mathbf{u}^+(\mathbf{x})$ and $\mathbf{u}^-(\mathbf{x})$ are continuous functions) such that sliding mode on the manifold $s(\mathbf{x}) = 0$ is enforced. As a result, the feedback system exhibits the desired dynamic behavior after the sliding mode starts.

In this paper, the design method rests upon Ackermann's formula and is oriented toward obtaining a discontinuity plane equation in explicit form as well in terms of the original system without transforming it into the phase canonic or regular forms. For a linear plant operating under uncertainty conditions, the control enforces sliding motions governed by a linear equation with the desired eigenvalue placement and independent of perturbations.

Two sliding mode control versions are studied. The first one implies that the sliding mode with the desired dynamics is described by a reduced-order system and arises after a finite-time interval needed for the state to reach a discontinuity surface. For the second control law a dynamic controller is designed such that the orders of the sliding mode equation and the original system coincide and sliding motion exists starting from any initial state. As a result, for the second version the system motion is insensitive with respect to disturbances starting from an initial time instant, while for the first version the motion preceding the sliding mode depends on disturbances.

II. REDUCED-ORDER SLIDING MODE

Consider a controllable system described by a differential equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}[u + f(\mathbf{x}, t)] \quad (2)$$

where $\mathbf{x}$ is an n -dimensional state vector, u is a scalar control, $\mathbf{A}$ and $\mathbf{b}$ are a known matrix and vector, $f(\mathbf{x}, t)$ is a nonlinear perturbation with known upper bound $f_0(\mathbf{x}, t)$

$$|f(\mathbf{x}, t)| < f_0(\mathbf{x}, t).$$

As it follows from (2), the control and disturbance vectors ($\mathbf{b}u$ and $\mathbf{b}f$) are colinear, so the matching condition [3] is satisfied, and then the sliding mode in any plane is insensitive with respect to the disturbance.

The design of sliding mode control in (2) implies a selection of a plane $s = \mathbf{c}^T \mathbf{x} = 0$ and then the control enforcing sliding mode in $s = 0$. The sliding mode equation is of the $(n-1)$ th order and does not depend on the perturbation. The desired dynamic properties may be provided by a proper choice of the vector $\mathbf{c}$. Traditionally, first the sliding mode equation is derived and then the conventional methods of the linear theory are applied.

The aim of this paper is to show how the vector $\mathbf{c}$ may be found in an explicit form without the sliding motion equation using Ackermann's formula, as far as the eigenvalue placement task is concerned.

The desired eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of the linear system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}u_a$ may be assigned using Ackermann's formula [1]

$$u_a = -\mathbf{k}^T \mathbf{x}, \quad \mathbf{k}^T = \mathbf{e}^T P(\mathbf{A}) \quad (3)$$

where

$$\mathbf{e}^T = (0, \dots, 0, 1)(\mathbf{b}, \mathbf{A}\mathbf{b}, \dots, \mathbf{A}^{n-1}\mathbf{b})^{-1}$$

$$P(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_{n-1})(\lambda - \lambda_n).$$

Suppose now that the real or pairwise conjugate complex values $\lambda_1, \dots, \lambda_{n-1}$ are the desired eigenvalues of the sliding mode.

Theorem¹: If

$$\mathbf{c}^T = \mathbf{e}^T P_1(\mathbf{A}) \quad (4)$$

¹The authors thank one of the reviewers who drew their attention to the simpler proof of the equality $\mathbf{c}^T \mathbf{b} = 1$, which let them simplify both the proof of this theorem and the control algorithm.

with $P_1(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_{n-2})(\lambda - \lambda_{n-1}) = p_1 + p_2\lambda + \cdots + p_{n-1}\lambda^{n-2} + \lambda^{n-1}$, then $\lambda_1, \dots, \lambda_{n-1}$ are the eigenvalues of the sliding mode dynamics in the plane $s = \mathbf{c}^T \mathbf{x} = 0$.

Proof: According to Ackermann's formula (3), $\lambda_1, \dots, \lambda_{n-1}, \lambda_n$ are eigenvalues of the matrix $\mathbf{A}^* = \mathbf{A} - \mathbf{b}\mathbf{k}^T$, with λ_n as an arbitrary value. Vector $\mathbf{c}^T$ is a left eigenvector of $\mathbf{A}^*$ corresponding to λ_n . Indeed, as follows from (3) and (4):

$$\mathbf{c}^T \mathbf{A}^* = \mathbf{c}^T \mathbf{A} - \mathbf{c}^T \mathbf{b} \mathbf{e}^T P(\mathbf{A}).$$

Since

$$\begin{aligned} \mathbf{c}^T \mathbf{b} &= \mathbf{e}^T P_1(\mathbf{A}) \mathbf{b} \\ &= (00 \cdots 01)(\mathbf{b}, \mathbf{A}\mathbf{b}, \dots, \mathbf{A}^{n-1}\mathbf{b})^{-1} \\ &\quad \times (\mathbf{b}, \mathbf{A}\mathbf{b}, \dots, \mathbf{A}^{n-1}\mathbf{b})(p_1 p_2 \cdots p_{n-1} 1)^T \\ &= 1 \end{aligned} \quad (5)$$

then

$$\mathbf{c}^T \mathbf{A}^* = \mathbf{c}^T \mathbf{A} - \mathbf{e}^T P_1(\mathbf{A})(\mathbf{A} - \lambda_n \mathbf{I})$$

which reduces with (4) to

$$\mathbf{c}^T \mathbf{A}^* = \lambda_n \mathbf{c}^T. \quad (6)$$

The system $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{b}\mathbf{k}^T)\mathbf{x} + \mathbf{b}[u - u_a + f(\mathbf{x}, t)]$ is now transformed such that $s = \mathbf{c}^T \mathbf{x}$ becomes the last state and the first $n-1$ states $\mathbf{x}^1 = [x_1, x_2, \dots, x_{n-1}]^T$ remain unchanged, i.e.,

$$\begin{bmatrix} \mathbf{x}^1 \\ s \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{c}^T \end{bmatrix} \mathbf{x} = \mathbf{T}\mathbf{x}.$$

For $\mathbf{T}$ to be invertible, the last component of $\mathbf{c}^T$ must be nonzero. Since this vector is nonzero, the condition can always be satisfied by reordering the components of the state vector $\mathbf{x}$. Under the conditions (5) and (6), the transformed system is

$$\dot{\mathbf{x}}^1 = \mathbf{A}_1 \mathbf{x}^1 + \mathbf{a}^s s + \mathbf{b}^1 [u - u_a + f(\mathbf{x}, t)] \quad (7)$$

$$\dot{s} = \lambda_n s + u - u_a + f(\mathbf{x}, t) \quad (8)$$

where

$$\begin{bmatrix} \mathbf{A}_1 & \mathbf{a}^s \\ \mathbf{0} & \lambda_n \end{bmatrix} = \mathbf{T}(\mathbf{A} - \mathbf{b}\mathbf{k}^T)\mathbf{T}^{-1}, \quad \begin{bmatrix} \mathbf{b}^1 \\ 1 \end{bmatrix} = \mathbf{T}\mathbf{b}.$$

The spectrum of the matrix $\mathbf{A}_1$ consists of the desired eigenvalues $\lambda_1, \dots, \lambda_{n-1}$.

To derive the sliding mode equation in the plane $s = 0$, the solution to the algebraic equation $\dot{s} = 0$ with respect to u should be substituted into (7). It results in the motion equation

$$\dot{\mathbf{x}}^1 = \mathbf{A}_1 \mathbf{x}^1 \quad (9)$$

with the desired dynamics, independent of the unknown perturbation $f(\mathbf{x}, t)$. This ends the proof.

The result has a transparent geometric interpretation. Vector $\mathbf{c}^T$ is a left eigenvector of the matrix $\mathbf{A}^*$ corresponding to the eigenvalue λ_n . It means that $\ker \mathbf{c}^T$ is an invariant subspace of $\mathbf{A}^*$ determined by the previously selected set of $(n-1)$ eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_{n-1}$. If the sliding mode is enforced in the plane $s = \mathbf{c}^T \mathbf{x} = 0$ (the second stage of the design procedure), then it exhibits the desired dynamics. Note that the design of the plane $s = \mathbf{c}^T \mathbf{x} = 0$ does not imply assigning the eigenvalue λ_n ; it appears in the proof of the theorem only and may take an arbitrary value.

The discontinuous control u is designed to enforce a sliding mode in the plane $s = 0$ (or to provide the conditions $(s\dot{s}) < 0$ and $\lim(s\dot{s}) < 0$ with $s \Rightarrow 0$)

$$\begin{aligned} \dot{s} &= \mathbf{c}^T \mathbf{A}\mathbf{x} + u + f(\mathbf{x}, t) \\ u &= -M(\mathbf{x}, t) \operatorname{sign} s \end{aligned} \quad (10)$$

where $M(\mathbf{x}, t)$ is chosen such that

$$M(\mathbf{x}, t) > |\mathbf{c}^T \mathbf{A}\mathbf{x}| + f_0(\mathbf{x}, t).$$

If the control may take only two extreme values $+M_0$ or $-M_0$ (which is common in applications), then (10) with $M(\mathbf{x}, t) = M_0$ enforces a sliding mode in the plane $s = 0$ governed by (9) as well. Of course, the domain of initial conditions and the perturbation should be bounded.

Example: Let $\lambda = -1$ be the desired eigenvalue of sliding motion for the second-order system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}[u + f(\mathbf{x}, t)]$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

According to (4)

$$\mathbf{c}^T = (0, 1)(\mathbf{b}, \mathbf{A}\mathbf{b})^{-1}P_1(\mathbf{A}), \quad P_1(\mathbf{A}) = \mathbf{A} + \mathbf{I}$$

$$\mathbf{c}^T = (-1, 2)$$

and the sliding surface equation is of the form

$$s = -x_1 + 2x_2 = 0$$

(note that $\mathbf{c}^T\mathbf{b} = 1$).

By the equivalent control method, the solution to the system $s = 0$, $\dot{s} = 0$ with respect to x_2 , and u should be substituted into the original system to derive the sliding motion equation

$$u_{eq} = -x_1 - f(x, t), \quad x_2 = \frac{1}{2}x_1$$

and

$$\dot{x}_1 = -x_1.$$

The sliding mode is determined by the eigenvalue $\lambda = -1$ and does not depend on the disturbance $f(x, t)$.

The *design procedure* is summarized as follows.

- Step 1) The desired spectrum of the sliding motion $\lambda_1, \dots, \lambda_{n-1}$ is selected.
- Step 2) The equation of the discontinuity plane $s = \mathbf{c}^T\mathbf{x} = 0$ is found as

$$\mathbf{c}^T = \mathbf{e}^T(\mathbf{A} - \lambda_1\mathbf{I})(\mathbf{A} - \lambda_2\mathbf{I}) \cdots (\mathbf{A} - \lambda_{n-1}\mathbf{I}).$$

- Step 3) The discontinuous control (10) is designed.

Remark: It follows from

$$\dot{s} = \mathbf{c}^T\mathbf{A}\mathbf{x} + [u + f(t)]$$

that sliding mode may be enforced in an unperturbed system ($f(t) \equiv 0$) if

$$u = -\omega\|\mathbf{x}\| \text{ sign } s$$

with some finite positive number ω . The control tends to zero in systems with asymptotically stable sliding modes.

III. FULL-ORDER SLIDING MODE

Control (10) provides the desired motion in the system after a finite-time interval needed for the state to reach the manifold $s = 0$. As a result, the system motion depends on the perturbation and is not governed by (9) within this interval. In this section, a design method is developed to enforce a sliding mode with the desired dynamics independent of the perturbation starting at any initial state. The motion is represented by a linear time-invariant homogeneous system of full order with the desired eigenvalues.

Suppose that the set $\lambda_1, \dots, \lambda_n$ is the desired spectrum of the feedback system and $u_a = -\mathbf{k}^T\mathbf{x}$ is the linear control providing this spectrum for the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}u_a$. According to Ackermann's formula, u_a may be found from (3).

Complement (2) by an additional first-order dynamic subsystem

$$\dot{z} = -\mathbf{b}^T\mathbf{A}\mathbf{x} - (\mathbf{b}^T\mathbf{b})u_a, \quad z(0) = -\mathbf{b}^T\mathbf{x}(0) \quad (11)$$

and let the control u in (2) be the discontinuous function of the states $\mathbf{x}$ and z

$$u = -M(\mathbf{x}, t) \text{ sign } s \quad (12)$$

$$s = \mathbf{b}^T\mathbf{x} + z. \quad (13)$$

The motion projection on the one-dimensional s -space is governed by

$$\dot{s} = (\mathbf{b}^T\mathbf{b})(u + f(\mathbf{x}, t) - u_a). \quad (14)$$

The condition for the existence of a sliding mode is of the form

$$M(\mathbf{x}, t) > |u_a| + f_0(\mathbf{x}, t). \quad (15)$$

The sliding mode arises at the initial instant because $s(0) = 0$. As before, the solution to the algebraic equation $\dot{s} = 0$ with respect to u should be substituted into (2) to obtain the sliding mode equation in the plane $s = 0$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}u_a.$$

The control u_a was designed in compliance with Ackermann's formula; therefore, the system behavior in the presence of an unknown perturbation $f(\mathbf{x}, t)$ is described by the linear time-invariant equation with the assigned spectrum.

If the control may take only two values M_0 and $-M_0$, the magnitude of control M_0 should satisfy (15), which means that initial conditions and the perturbation should be bounded.

Use of the additional dynamic subsystem with the specific initial condition (11) enforces a sliding mode with the desired dynamics of order equal to that of the original system. The full-order sliding motion exists for any $t \geq 0$.

Design Procedure:

- Step 1) Choose the desired spectrum of the feedback system $\lambda_1, \dots, \lambda_n$.
- Step 2) Linear control $u_a = -\mathbf{k}^T\mathbf{x}$ is found in accordance with Ackermann's formula $\mathbf{k}^T = \mathbf{e}^T P(\mathbf{A})$.
- Step 3) The dynamic part of the controller is designed $\dot{z} = -\mathbf{b}^T\mathbf{A}\mathbf{x} - (\mathbf{b}^T\mathbf{b})u_a$, $z(0) = -\mathbf{b}^T\mathbf{x}(0)$.
- Step 4) The equation of the discontinuity plane $s = 0$ is found $s = \mathbf{b}^T\mathbf{x} + z$.
- Step 5) The discontinuous control (12), (13), (15) is designed.

IV. CONCLUSION

The proposed methods enable the sliding mode control design, based on Ackermann's formula, in terms of the original system rather than in terms of the sliding mode equations. The system with a static controller exhibits the desired dynamics and insensitivity with respect to unknown perturbations after a finite-time interval, while the system with a dynamic controller has the same properties starting from any initial state.

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An Algebraic Criterion for Robust Stability of Linear Control Systems

Boris E. Dainson and Daniel R. Lewin

Abstract—A new robust stability test for linear control systems is described. The condition at which robust stability is violated is transformed into an equivalent problem in which the existence of a real root of a multivariable polynomial is investigated. This multivariable problem is reduced to that of the solvability of a set of univariable polynomial equations in real numbers, for which a number of efficient numerical methods are available. The use of the method is illustrated in the design of feedback control for an open-loop unstable batch chemical reactor.

Index Terms—Linear systems, multivariable polynomials, robust control, robust performance, robust stability.

NOMENCLATURE

$c(s)$	Feedback controller transfer function.
k	Process gain.
$p(s)$	True plant transfer function.
$\tilde{p}(s)$	Nominal plant model transfer function.
P, Q	Multivariable polynomials.
$r(s)$	Reference signal.
T	Time constant.
$w(s)$	Scalar performance weight.
Δ	Uncertainty matrix.
$\tilde{\epsilon}(s)$	Nominal sensitivity function, $\tilde{\epsilon} = (\tilde{p}c + 1)^{-1}$.
λ	Tuning parameter of the IMC controller.
$\tilde{\eta}(s)$	Nominal complementary sensitivity function, $\tilde{\eta} = \tilde{p}c(\tilde{p}c + 1)^{-1}$.
$\mu(M)$	Structured singular value of M .
ω	Frequency.

I. INTRODUCTION

An accepted procedure for analyzing the robustness of multi-input/multi-output (MIMO) control systems is the structured singular value approach [6]. In this procedure, the structured singular value (μ) is computed at each frequency with respect to the structure of the perturbation matrix Δ in the standard M - Δ form. For robust performance, the M matrix contains information on the nominal control system as well as weights representing process uncertainty and closed-loop performance specifications, while the Δ matrix consists of bounded perturbations which define the process uncertainty (Δ_u) and measures of desired performance (Δ_p). The elimination of the

Δ_p perturbation from the analysis reduces the M - Δ structure to that of defining the robust stability problem. If $\|\Delta\| \leq 1$, then closed-loop stability or performance for the perturbed M - Δ structure can be guaranteed if and only if

$$\mu(M(j\omega)) < 1 \quad \forall \omega. \quad (1)$$

Extensions of the original approach by Fan *et al.* [8] can treat the case where the uncertainty matrix is block diagonal with complex and real blocks. This method generates upper bounds on μ through the numerical solution of a convex optimization problem. In many cases, the uncertainty matrix can be transformed into a diagonal form. For uncertainty defined as bounded real perturbations, mapping methods developed by Saeki [11] were successfully implemented by de Gaston and Safonov [7] for the exact solution of the stability problem. The procedure converges iteratively to the perturbation Δ which causes instability. For a real diagonal perturbation matrix, a relatively simple procedure to compute the upper bound on μ was presented by Dailey [5], determined by a specific constraint on the values of uncertainty elements. The procedure is relatively simple and returns the value of the destabilizing perturbation.

The disadvantages of all these methods are that they either provide only sufficient conditions for the system stability or require extensive iterative computations. The approach proposed here examines the system stability as the problem of existence of real roots of multivariable polynomial in a region defined by the structure of the perturbation matrix Δ . The existence of real roots of this polynomial is then equivalent to the instability of the control system. The advantage of such an approach is that it gives an exact, necessary, and sufficient stability condition.

The problem of existence of real solutions of a multivariable polynomial with unconstrained or constrained real variables appears in many systems theory problems. Among these are the existence of stationary and periodic solutions of nonlinear systems, tests for Lyapunov stability, multidimensional stability, and robust stability tests. These problems can be formulated in terms of positivity tests relying on continuity arguments and can be solved by elementary decision methods. A thorough survey of elementary decision methods has been presented by Bose [4]. An introduction to these methods with application to the solution of the feedback stabilization problem can be found in Seidenberg [12] and Anderson *et al.* [1]. These methods perform a successive reduction of the number of variables until one is left with a finite set of univariable polynomial equations, which can then be treated by the standard Sturm procedure. As a result, the original problem is solved by a finite number of rational operations. Both these approaches treat inequality constraints as equality constraints by introducing additional slack variables. When applied to robustness analysis with bounded real perturbations, these approaches will significantly increase the number of unknown variables that need to be solved. The approach proposed here, which is similar to that presented by Tarski [13] does not have this disadvantage.

The problem of existence of real solutions of a multivariable polynomial was also considered by Bikart and Jury [3] and by Walach and Zeheb [14]. The proposed approach is similar to the latter but can be shown to lead to the polynomials of lower degree. Nevertheless, the degree of the polynomials to be treated still remains to be a fast growing function of the number of uncertain parameters. This makes the approach effective only for problems with relatively low number of uncertainties.

II. PROBLEM FORMULATION

The linear system described by a stable transfer matrix $M(s)$ under the feedback perturbation Δ is stable if and only if the characteristic

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The authors are with the Department of Chemical Engineering, Technion I.I.T., Haifa 3200, Israel.

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